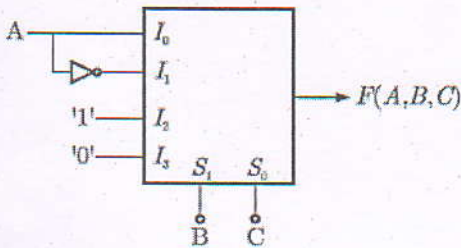




Semester: July 2025 – November 2025		
Maximum Marks: 50	Examination: End-Semester Examination	Duration: 2 Hrs.
Programme code: 03 Programme: BTech EXTC	Class: TY BTech EXTC	Semester: V (SVU 2023)
Institute/School/Department: K. J. Somaiya School of Engineering	Name of the department: EXTC	
Course Code: 216U03C303	Name of the Course: Digital Logic Design	
Instructions: 1) Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question Statement	Max. Marks
Q.1	Attempt any two	10
i)	Convert a) Decimal (168) to Hex and Octal System b) Decimal (57) to grey code	05
ii)	The figure below shows implementation of a 3 variable logic function using 4:1 multiplexer. Write truth table and the minterm expression represented in the figure 	05
iii)	If $Y(A,B,C) = \prod M(0,1,4,5,7)$ write minimised POS expression. Realise using NOR gates	05
Q.2	Attempt any one	10
i)	Reduce the following expression using Quine-McCluskey method $F(A,B,C,D) = \sum m(0,1,3,9,11,12,14,15)$	10
ii)	Realise $f1 = \sum m(0,2,4,5,8,10,12)$ and $f2 = \sum m(1,2,5,7,9,11,15)$ using suitable decoder circuit. Assume decoder has active low output.	10
Q.3	Attempt any one	10
i)	Draw circuit diagram of a 4 bit Johnson counter (Twisted Ring counter) using D flip flops along with truth table	10
ii)	Design a Synchronous counter that goes through sequence $0 \rightarrow 2 \rightarrow 5 \rightarrow 6 \rightarrow 7 \rightarrow 0$ Using JK flip flops	10
Q.4	Attempt the following	10
	Write truth tables and excitation tables of S-R , J-K and D flip flops. Explain what is race around condition in JK flip flop.	10



Q.5	Attempt the following	10
A)	Explain any one technique of Analog to Digital conversion (ADC) with circuit diagram . <u>OR</u>	5
A)	Compare Moore and Melay machines with block diagrams	5
B)	Define following terms related to logic families 1. Propagation Delay 2. Fan out 3. Noise Margin	5